

Sanyam Sharma

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WORK EXPERIENCE

- Indian Institute of Technology, Hyderabad** Hyderabad, Telangana
Research Intern Jan 2026 – Present
- Developed an AI-driven network automation system translating natural language intents into SONiC configurations
 - Integrated Batfish verification and GNS3 digital-twin testing for automated network change validation
 - Built GNS3 REST API-driven workflows for automated network topology discovery and configuration analysis in SONiC environments
 - Created Terraform code-generation datasets and benchmarked SLMs using CodeBLEU, CodeBERTScore and Pass@k metrics
- Indian Institute of Technology, Mandi** Mandi, Himachal Pradesh
Summer Intern Jun 2025 – Jul 2025
- Generated speech embeddings using MFCC, wav2vec2.0 and wav2vec2-phoneme for language identification
 - Implemented a Conformer-based PyTorch architecture achieving 79% accuracy on unseen multilingual speech samples
 - Developed a PyQt6 desktop GUI integrating pretrained AASIST and RawNet models for spoofed speech detection
- Indian Institute of Technology, Mandi** Mandi, Himachal Pradesh
Research Intern Jun 2024 – Jun 2024
- Set up data pipeline using Telegraf and InfluxDB to collect and store sensor data from an ESP32 via MQTT.
 - Deployed the system on a Raspberry Pi using Docker for real-time data processing.

EDUCATION

- Jawaharlal Nehru Government Engineering College** Mandi, Himachal Pradesh
B.Tech – CSE (AI & ML) 2022 – 2026
- Government Senior Secondary School, Kangoo** Hamirpur, Himachal Pradesh
Senior Secondary (Non-Medical) 2022

PROJECTS

- Transformer Serve | Live Demo | GitHub | Docker** 2026
- Tech:** Python, FastAPI, Docker, PyTorch, Hugging Face Transformers
- Built a configurable inference API serving any Hugging Face causal or seq2seq text-generation model, with models swappable via environment variables and no code changes required
 - Containerized with Docker (CPU-optimized PyTorch build) and deployed a live public API on Hugging Face Spaces
- AI-Powered Assistive Vision for Blind People - Major Project** Aug 2025 – Dec 2025
- Tech:** gTTS, FaceNet, Gemini-API, pytesseract, RaspberryPi
- Developed an AI-powered assistive vision smart cap for blind individuals using Raspberry Pi, camera, LiDAR, audio module
 - Implemented OCR-based text recognition to read printed text and deliver real-time audio output
 - Integrated obstacle detection with audio-based navigation for safe mobility
- Extension for malicious QR code and URL detection - Capstone Project** Jan 2025 – May 2025
- Tech:** Python, Flask, Scikit-Learn, JavaScript
- Engineered features from large-scale malicious and legitimate URL/QR datasets
 - Built a stacked ensemble ML model (5 base + meta-classifier), achieving 91.3% accuracy
 - Developed a Chrome extension with Flask backend for real-time malicious URL and QR code detection
- SpoofedSpeechGUI | GitHub** Jun 2025 - July 2025
- Tech:** Python, PyQt6, PyTorch, Librosa
- Extracted audio features using Librosa and applied pretrained AASIST and RawNet models for spoofed speech detection
 - Developed a PyQt6 desktop GUI application for real-time inference

TECHNICAL SKILLS

Languages: Python, C++, SQL, HTML, CSS
Libraries: PyTorch, FastAPI, Scikit-learn, TensorFlow (Keras), NumPy, Pandas, Flask
Tools: Docker, Git, GitHub, Linux, Ollama, Postman, Ansible, Terraform, GNS3, Batfish

PUBLICATIONS

- HybridStack-MLP: Advanced Ensemble Learning for Malicious QR Code and URL Detection** 2025
- Proc. 2nd IEEE International Conference for Women in Computing (InCoWoCo 2025)*, IEEE Xplore, 2025.
[doi:10.1109/InCoWoCo68239.2025.11407094](https://doi.org/10.1109/InCoWoCo68239.2025.11407094)

ACHIEVEMENTS

- Winner — State Level Smart Hackathon (SLSH), Himachal Pradesh** 2025